

BILKENT UNIVERSITY

PHYS 101 Lab Project Report:

Investigation of Translational and

Rotational Dynamics



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Physics 101- Section 11

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PHYS 101 EXPERIMENT: INVESTIGATION OF TRANSLATIONAL AND ROTATIONAL DYNAMICS

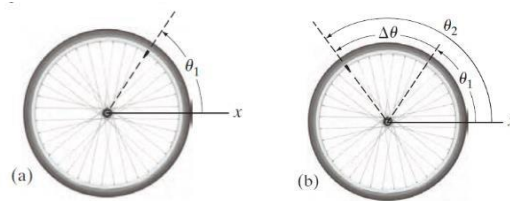
Objective:

The aim of this experiment is to observe and compare the landing times and landing speeds of an object rolling without slipping on an inclined plane where there is almost no friction, and an object sliding down without rotating.

Introduction:

Angular quantities like angular acceleration and angular velocity define rotational motion. Since these quantities are defined by analogy with the comparable quantities in linear motion and characterize the item as a whole, they are the same at every location in a rotating rigid object. Additionally, the item has acceleration and translational velocity at each location, however these differ from one location to another.

An object's angular displacement when it rotates from a starting location determined by θ_1 to a finishing position determined by θ_2 is



(Giancoli 266)

$$\Delta\theta = \theta_2 - \theta_1$$

The rate at which an object's angular location changes in relation to time is known as its average angular velocity when it rotates around a fixed axis:

$$\bar{\omega} = \frac{\Delta\theta}{\Delta t}$$

where $\Delta\theta$ is the rotational angle of the item during the time interval t . The limit of this ratio as Δt gets closer to 0 is **the instantaneous angular velocity**:

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

If there is constant acceleration, the average speed is half the sum of the initial speed and the final speed.

$$\bar{v} = \frac{(v + u)}{2}$$

“Angular velocity (denoted by ω , the Greek lowercase letter omega) is defined by analogy with linear (translational) velocity (v)” (Giancoli 266)

$$\mathbf{v} = \mathbf{r} \times \boldsymbol{\omega}$$

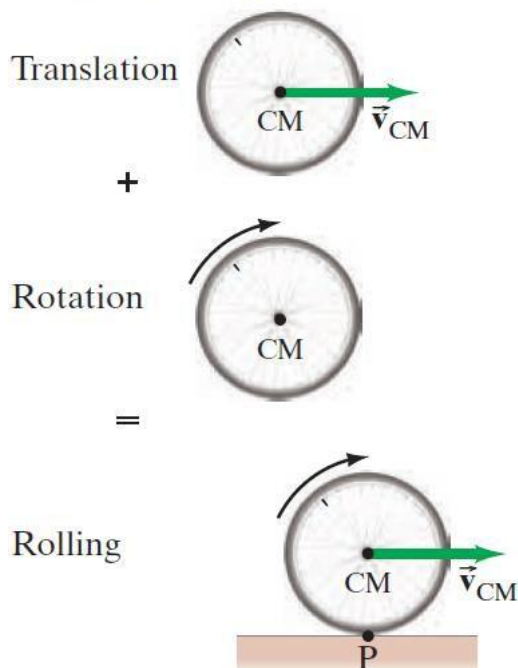
	Linear	Rotational
Position	x	θ
Velocity	$v = \frac{dx}{dt}$	$\omega = \frac{d\theta}{dt}$

(Lumen Learning 483)

The moment of inertia can be used to measure rotational inertia, while mass can be used to measure linear inertia. An object's inertia and resistance to changes in linear velocity rise with size, much like translational motion. A stiff substance or system of particles' resistance to variations in angular velocity at a fixed axis of rotation is likewise increased by its moment of inertia.

$$I = \sum_j m_j r_j^2$$

 A wheel rolling without slipping can be considered as translation of the wheel as a whole with velocity $\vec{v}_{CM}$ plus rotation about the CM.



(Giancoli 284)

A rotating object will possess both of them translational and rotational kinetic energy when the center of mass (cm) is in translational motion.

$$K = \frac{1}{2} I \omega^2$$

, which, in the case of a fixed axis of rotation, provides the rotational kinetic energy. The object will also have translational kinetic energy if it is in motion, like a wheel rolling on the ground. We observe that the rolling item is undergoing pure rotation about the contact point P in order to calculate the total kinetic energy. The speed of a point on the edge of the wheel with respect to its center is equal to the speed V of the center with respect to the ground. Both speeds are related to the radius R by $V = R\omega$.

Therefore, the angular velocity ω about point P is the same ω for the wheel rotating about its center. That is, the total kinetic energy about point P is pure rotational,

$$K_{\text{tot}} = \frac{1}{2} I_P \omega^2$$

, where I_P is the moment of inertia of the rolling object about the instantaneous axis at P.

Using parallel axis theorem, it can be written K_{tot} in terms of the center of mass:

$$I_P = I_{\text{CM}} + MR^2$$

Thus,

$$K_{\text{tot}} = \frac{1}{2} I_{\text{CM}} \omega^2 + \frac{1}{2} MR^2 \omega^2$$

but $R\omega = v_{\text{CM}}$, the speed of the center of mass. Thus, a rolling object's total kinetic energy equals

$$K_{\text{tot}} = \frac{1}{2} I_{\text{CM}} \omega^2 + \frac{1}{2} M v_{\text{CM}}^2$$

, where v_{CM} is the linear velocity of CM, I_{CM} is the moment of inertia about an axis passing through CM, ω is the angular velocity about this axis, and M is the object's total mass.

Hence, if we wanted to separate the kinetic energies and see which one came from where, the result would look like this:

Rotational	Translational
$I = \sum_j m_j r_j^2$	m
$K = \frac{1}{2} I \omega^2$	$K = \frac{1}{2} m v^2$

Gravitational potential energy is one of the most widely recognized types of potential energy.

“At any vertical height ‘y’ above a reference point (the coordinate system's origin),
the gravitational potential energy, U, can be defined as

$$U_{\text{grav}} = mgy$$

The force of gravity between the Earth and mass m is linked to potential energy.

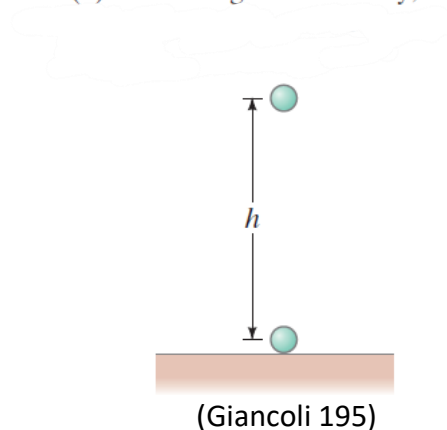
Therefore, U_{grav} denotes the mass's gravitational potential energy - Earth system, not just mass m” (Giancoli 197)

Conservative and non-conservative forces are the two categories of forces. Any force is referred to as conservative if its work on an object moving from one place to another is solely based on the object's starting and ending places and is unaffected by the specific route taken. Only position can determine a conservative force; it cannot be influenced by time or speed. It is simple to demonstrate the conservative nature of the gravitational force. An object with mass m close to the Earth's surface is subject to the gravitational force of the Earth, which is

$$\vec{F} = m\vec{g}$$

, where $\vec{g}$ is constant. This gravitational force's work on an object falling a vertical distance h is $W_G = Fd = mgh$ (see Fig. 8-1a)

FIGURE 8-1 Object of mass m :
(a) falls a height h vertically;



The quantity E which is called the **total mechanical energy of the system**, is the sum of the kinetic energy and potential energy of the system at any moment.

$$E = K + U = \frac{1}{2}mv^2 + mgy$$

As long as no nonconservative forces are in action, the total mechanical energy E stays constant. The total, $K + U$, remains constant:

When only conservative forces are involved, a system's total mechanical energy does not change during any process. It is maintained and stays the same.

For conservative forces, this is known as **the concept of conservation of mechanical energy**.

Equipments:

- An air table with its accessories
- A ruler
- Calculator
- Disc-shaped pulley
- String
- Holder
- Spark Timer

Procedure:

Part A: Calculating the magnitude of the final speed of the puck to be attached to the pulley and calculating how much time it takes to get from the top to the bottom

- 1.** To give the air table the desired slope, carefully place the wooden cube under the air table. The number written on the cube indicates the sine value of the slope angle.
- 2.** Carbon paper is laid on the air table. However, before laying the data paper on the carbon paper, check whether there is any obstacle or accumulation of dust.
- 3.** Since you'll only be utilizing one puck, put the other in a frame corner and fold the paper over to keep it from sliding. However, for both pucks to spark and produce dots on the data paper, they must be appropriately above the carbon paper.
- 4.** Set the timer to the intended value. The spark timer has different spark frequencies that you can choose from before starting an experiment. You can also evaluate your data and switch to a more suitable frequency.

5. Attach the pulley securely above the table and pass the holder to the puck, tie these two together with the help of string and wrap the string around the pulley.
6. In order to prevent oscillation of the puck while falling down, keep the pulley fixed and press the P switch. Besides, it makes you more confident about the setup by using a switch.
7. Place the two switches on top of each other and set the timer, then press both switches. Thus, the puck will make both translational and rotational movements until the ground level. Assume acceleration due to gravity as 9.8. Air resistance and friction are negligible.
8. A permanent record of the two pucks' subsequent positions at evenly spaced intervals of time is created by recording the spark positions as black prints on a data paper spread out on the air table. Record the displacement values in Table 1 after you start measuring it (not the distance between the two sites). You must fill out the entire table.
9. Using the values in Table 1, you can find the final speed using the last two points. Furthermore, you have to make a plot of x vs. t at Plot 1.
10. Another method is to find the final speed using the conservation of the mechanical energy.
11. You need to write the final speeds you found with two different methods in their places in Table 2 and calculate the percentage error value.

Part B: Calculating the magnitude of the final velocity of the free puck and calculating how much time it takes to get from the top to the bottom

1. Now place the free puck at the same height as the puck in Part A.
2. When ready, activate the pedals. The puck will slide down without rotating until it reaches the bottom.
3. Start to take your measurements and fill them in the blanks in Table 3.
4. Find the final speed using the values in Table 3. For this, there are two methods as mentioned in Part A. You should write the values you find with both of them in Table 4. Thus, you can find the error value.
5. You have to make a plot of x vs. t at Plot 2. Finally, you have to compare both the magnitude of the final velocity and the time that passed to reach the ground of both Part A and Part B and put these data into the Table 5.

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Data Point	Time t (s)	Displacement $x(t)$ (m)
1	5×10^{-2}	0.8×10^{-2}
2	10×10^{-2}	1.7×10^{-2}
3	15×10^{-2}	2.9×10^{-2}
4	20×10^{-2}	4.2×10^{-2}
5	25×10^{-2}	5.6×10^{-2}
6	30×10^{-2}	7.2×10^{-2}
7	35×10^{-2}	9.0×10^{-2}
8	40×10^{-2}	10.9×10^{-2}
9	45×10^{-2}	13×10^{-2}
10	50×10^{-2}	15.2×10^{-2}

Table 1

Magnitude of Final Velocity (m/s) (from the graph)	Magnitude of Final Velocity (m/s) (conservation of mechanical energy)	% Error
0.44	0.50	12

Table 2

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Data Point	Time t (s)	Displacement $x(t)$ (m)
1	5×10^{-2}	0.9×10^{-2}
2	10×10^{-2}	1.9×10^{-2}
3	15×10^{-2}	3.1×10^{-2}
4	20×10^{-2}	4.6×10^{-2}
5	25×10^{-2}	6.2×10^{-2}
6	30×10^{-2}	7.9×10^{-2}
7	35×10^{-2}	9.9×10^{-2}
8	40×10^{-2}	12×10^{-2}
9	45×10^{-2}	14.3×10^{-2}
10	50×10^{-2}	16.8×10^{-2}

Table 3

Magnitude of Final Velocity (m/s) (from the graph)	Magnitude of Final Velocity (m/s) (conservation of mechanical energy)	% Error
0.50	0.60	16.6

Table 4

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DYNAMICS

	Time taken to reach the ground (s)	Magnitude of Final Velocity (when object reached the ground) (m/s)
For Part A	95×10^{-2}	0.70
For Part B	90×10^{-2}	0.80

Table 5

Conclusion:

The aim of this experiment was to investigate translational and rotational dynamics by comparing motion with and without rotation.

In the rotational case, the puck exhibited slower acceleration, taking longer to reach the ground and achieving a lower final velocity compared to the non-rotational case. This was evident from the displacement vs. time graphs and the tabulated final velocities and times.

Error percentages were calculated as 12% for the rotational case and 16.6% for the non-rotational case. These percentages were determined by comparing the experimental final velocities with the theoretical final velocities derived from conservation of mechanical energy. Specifically:

In the rotational case, magnitude of the experimental final velocity was 0.44 m/s, while the theoretical value was 0.50 m/s, resulting in a 12% error.

In the non-rotational case, magnitude of the experimental final velocity was 0.50 m/s, while the theoretical value was 0.60 m/s, leading to a 16.6% error.

The errors in the both experiments can be attributed to factors such as friction, mistakes due to the person carrying out the experiments, and potential inaccuracies in equipment calibrations. These factors affected the precision of displacement and velocity measurements.

The displacement vs. time plots confirmed the expected parabolic shape, consistent with uniformly accelerated motion. This observation aligns with theoretical principles, with the rotational motion demonstrating the influence of rotational inertia.

Overall, the experiment successfully highlighted the differences between translational and rotational dynamics and their practical implications.

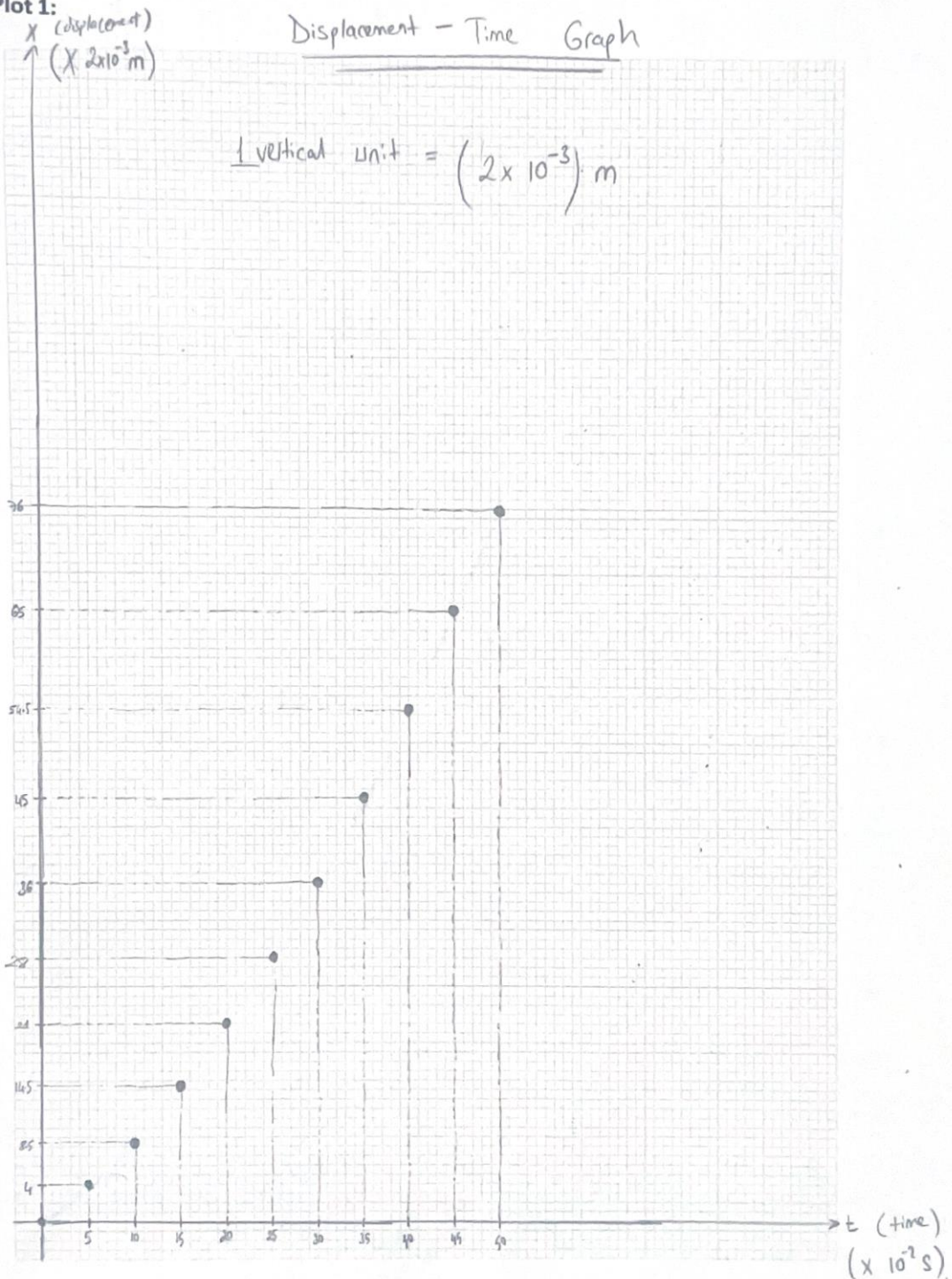
Plot 1:

NOTE:

$$X(t) = \overset{0}{x_0} + \overset{0}{u_0} \cdot t + \frac{1}{2} \cdot \underset{\substack{\uparrow \\ \text{(constant)}}}{a} \cdot t^2$$

then, we'll see
a parabola

Plot 1:



1 horizontal unit = $(1 \times 10^{-2}) \text{ s}$

Plot 2:

NOTE :

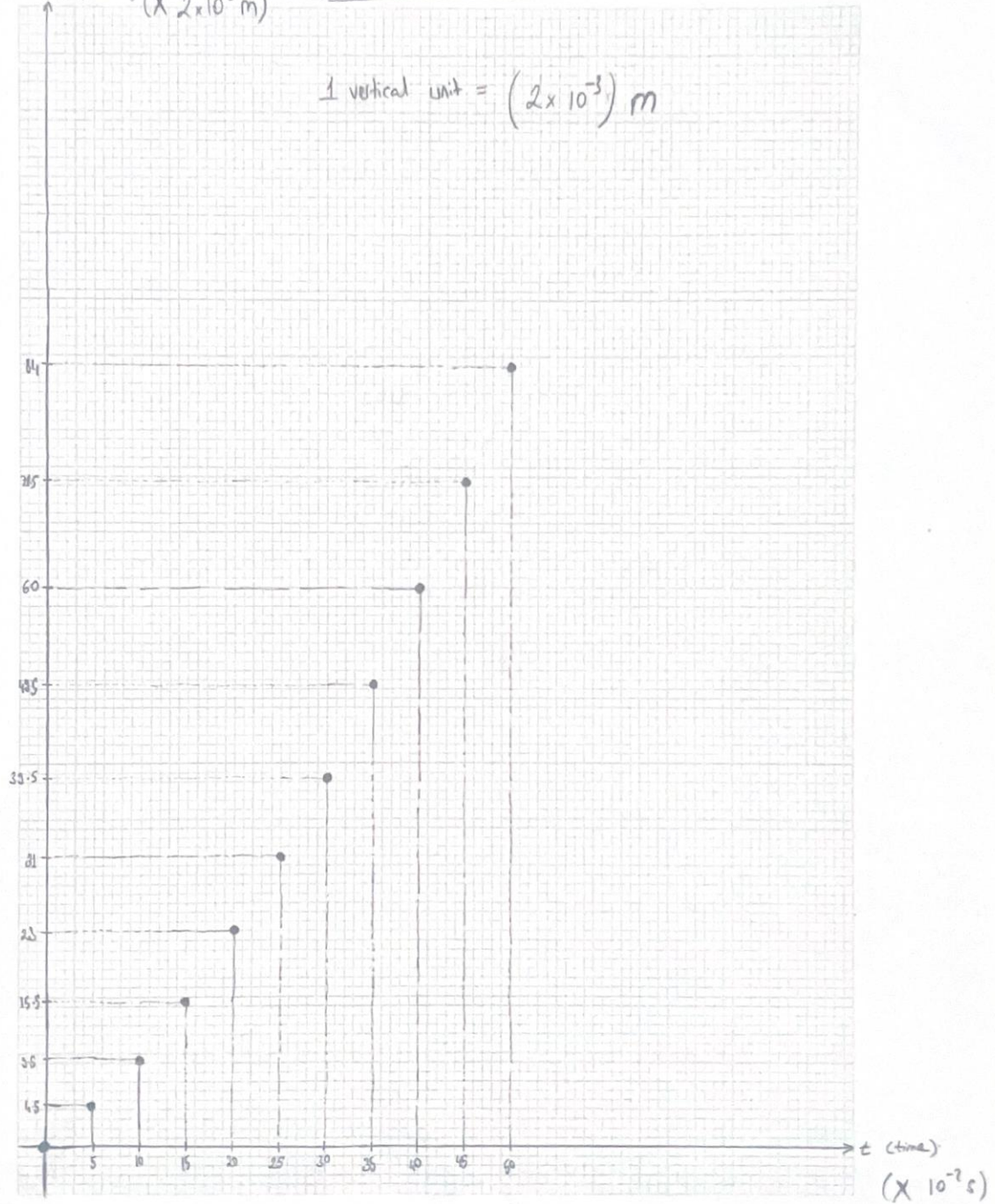
$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2 \quad \text{that, we'll see a parabola}$$

↑
(constant)

Plot 2:

x (displacement) ($\times 10^{-3}$ m) Displacement - Time Graph

1 vertical unit = (2×10^{-3}) m



1 horizontal unit = (1×10^{-2}) s

A youtube link to watch my experiment video:

<https://youtu.be/DSJZEYy2M2s?si=umWkkukiOxFJT5LC>

Sources:

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Giancoli, D. C. (2004). *Physics for scientists and engineers* (5th ed.). Pearson Educati